

Gravity Books Data Ware House Documentation

- Conceptual Data Modelling

There is an issue in the gravity books store analysis team that they can't do their work smoothly because of the lack of source of data to do the analysis on. Therefore, the proposed solution was creating a data mart for the sales modules so they can keep track of the sales, and orders delivery process and do the needed reports in time.

The reporters that the business asks for are to get the total order price and total fee (order price + shipping cost) per month, per shipping method, per author and per country.

And also the business wants to keep track of average days of process and days to deliver per shipping method. To achieve this business needs

- Logical Data Modelling

To achieve what businesses want the most suitable Architecture for this project is **Kimball architecture**. Due to the need for only one data mart for the sales module with two fact tables to get the different types of analysis.

The structure of the data mart will be a **Galaxy Schema with two fact tables**:

- Transactional fact: to track sales
- Accumulating snapshot fact: to track delivery process

The types of dimensions used will be:

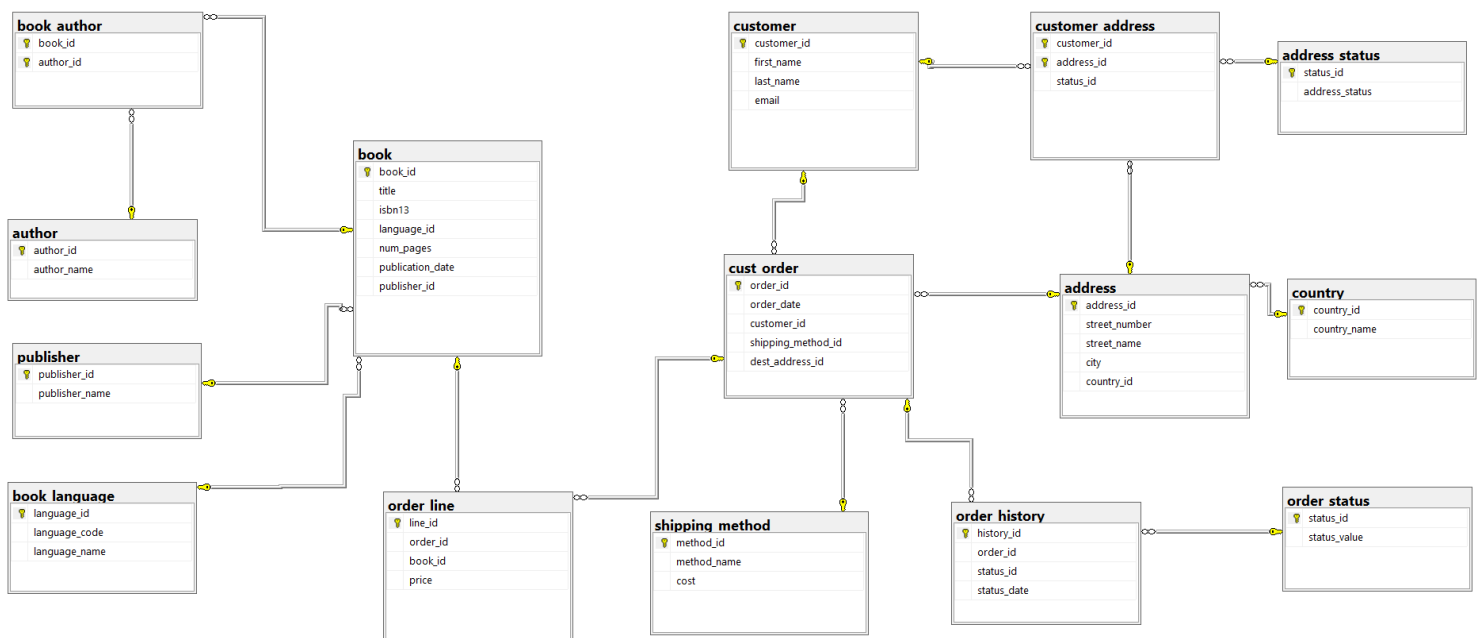
- Conformed Dimension for Date_Dim
- Degenerate Dimension for OrderID, LineID and HistoryID
- Junk Dimension for Status_Dim
- Snow flaked Dimension for Customer to Address and Book to Author as (many to many)
- Role Playing Dimension for Date_Dim with the Accumulating snapshot fact table
- Slowly Changing Dimension Type2 for Shipping_Dim to track changes in shipping cost

After integrating the data into the data mart we build a cube for much faster analysis.

first cube for sales with total fee and total order price as measures and the need dimensions based on business requirement are month, quarter shipping method, author and country.

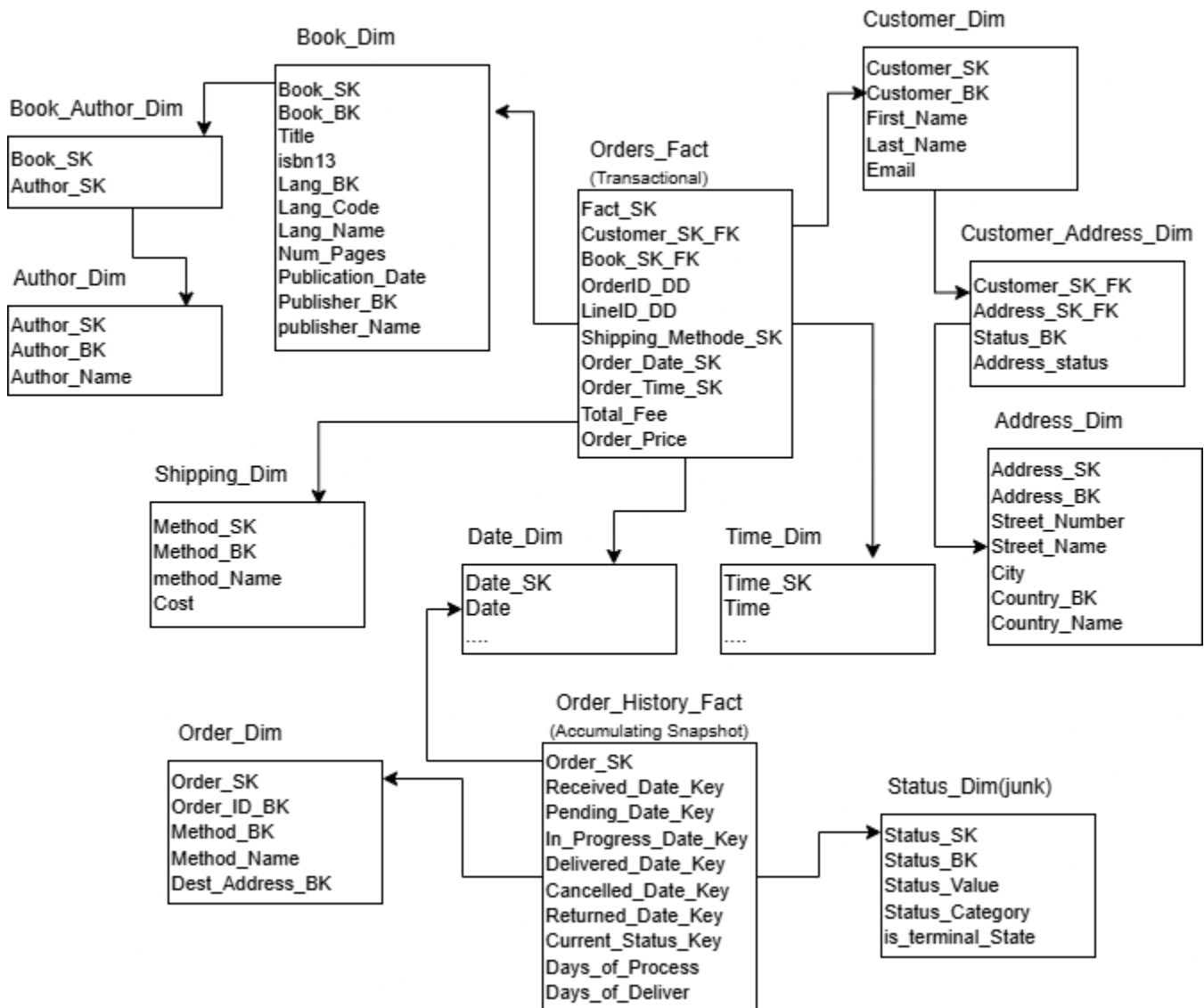
And the second cube with average days of process and average days to deliver as measures and shipping method dimension.

This is the OLTP Diagram:



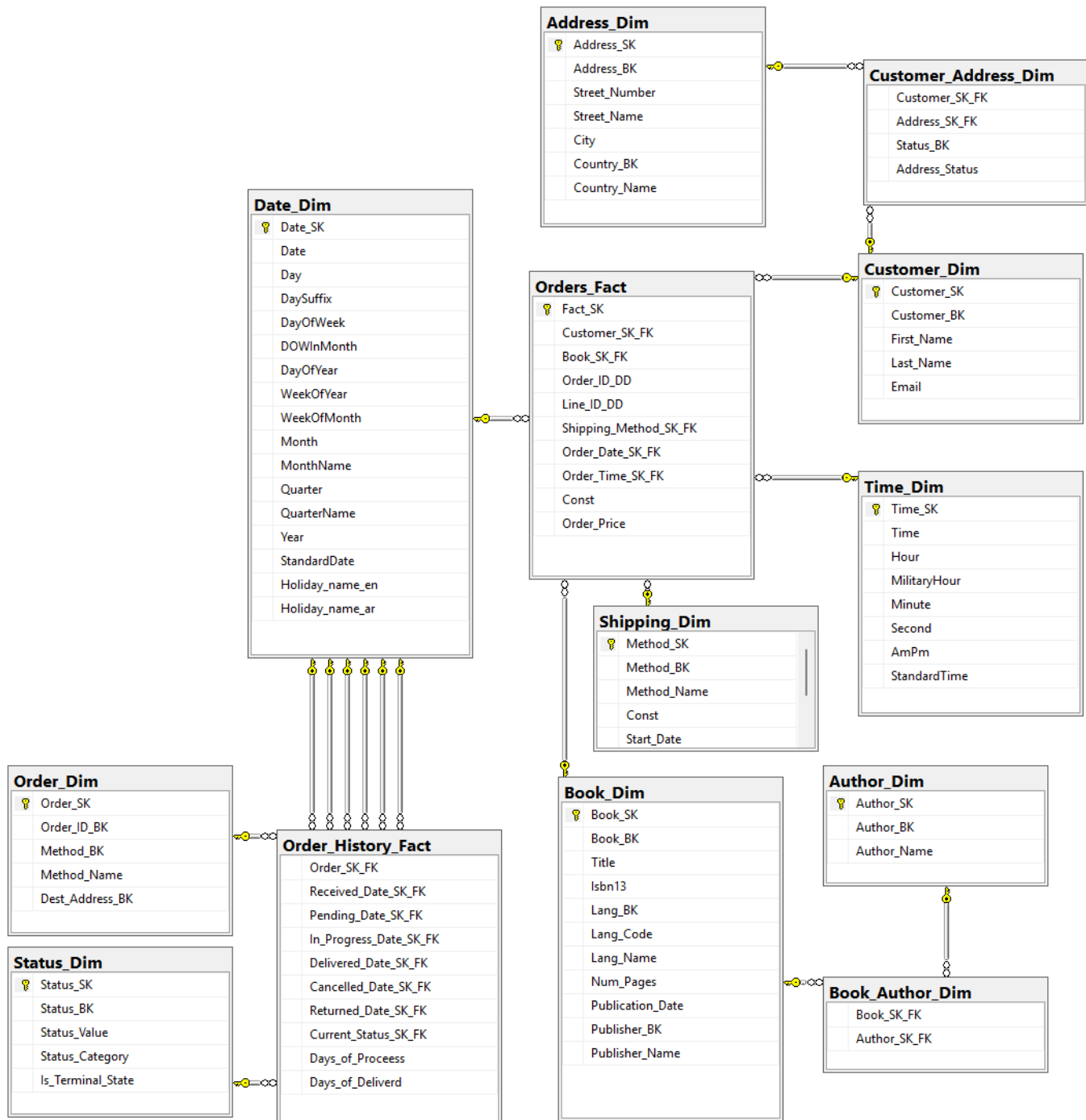
OLAP Solution:

GBs DWH Data Governance Sheet



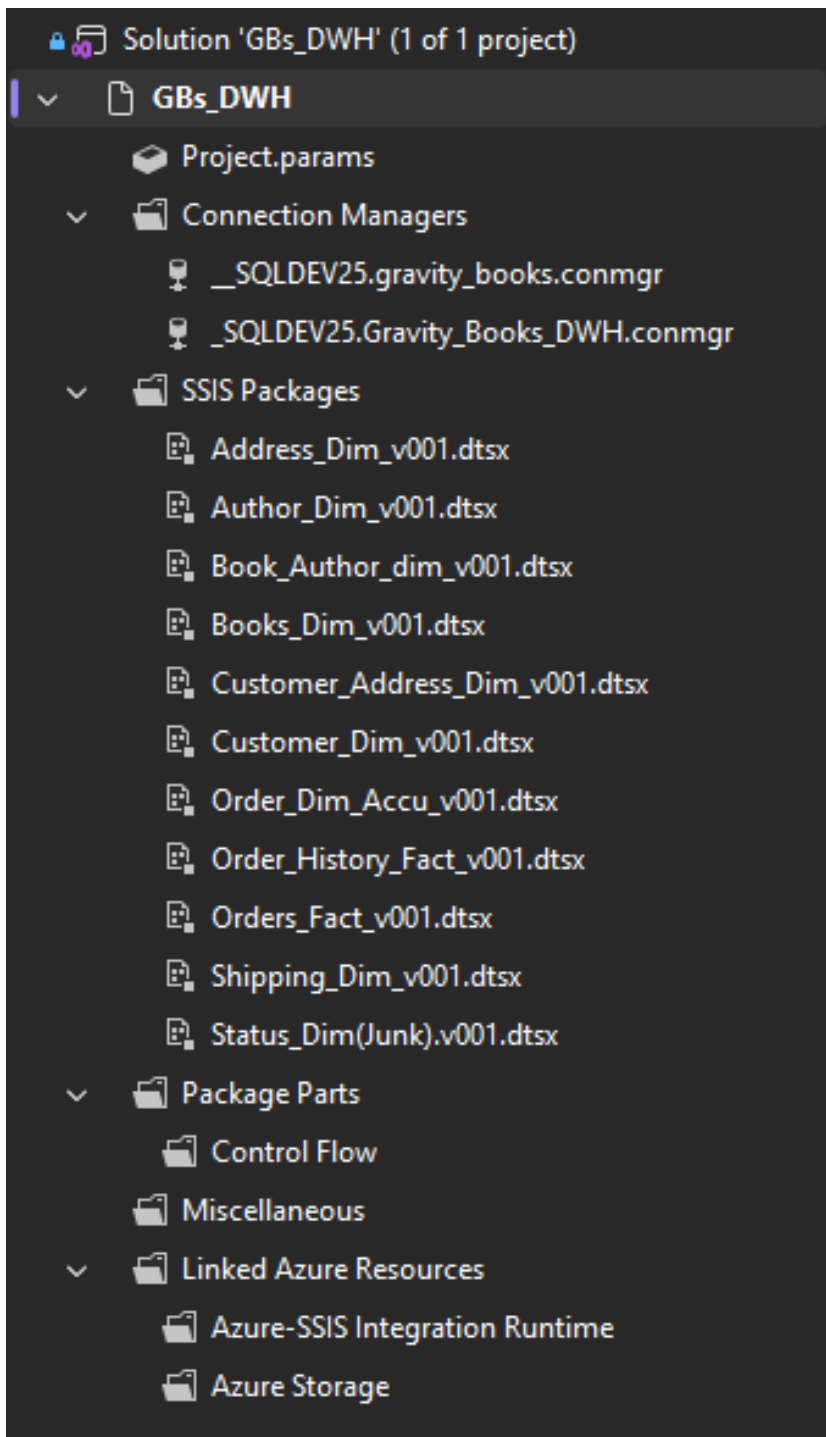
- Physical Data Modelling

The physical implementation of the solution:

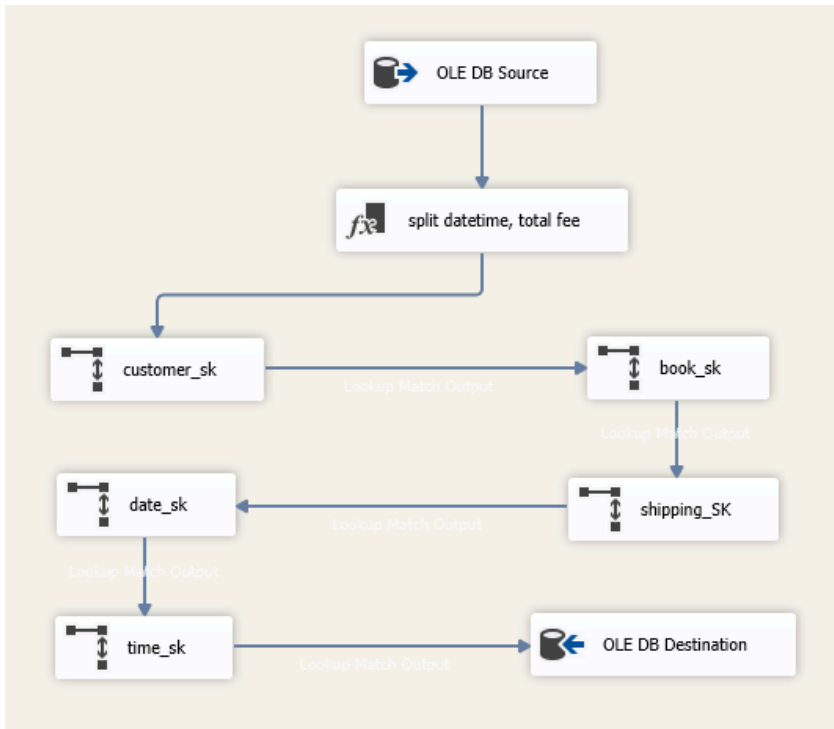


Then the data integrated from OLTP to OLAP using SSIS:

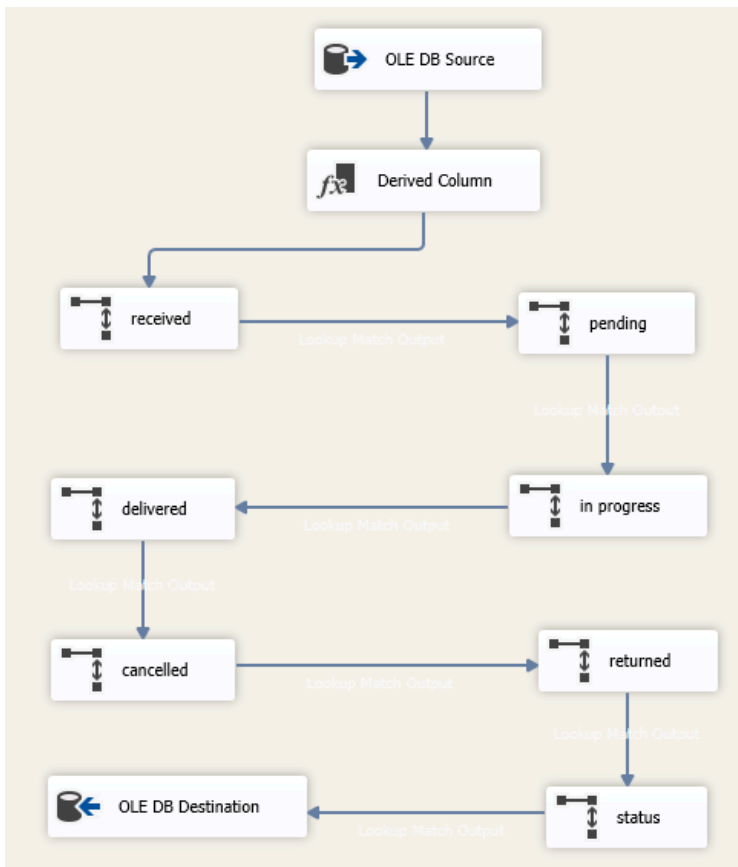
Creating package for each table integration:



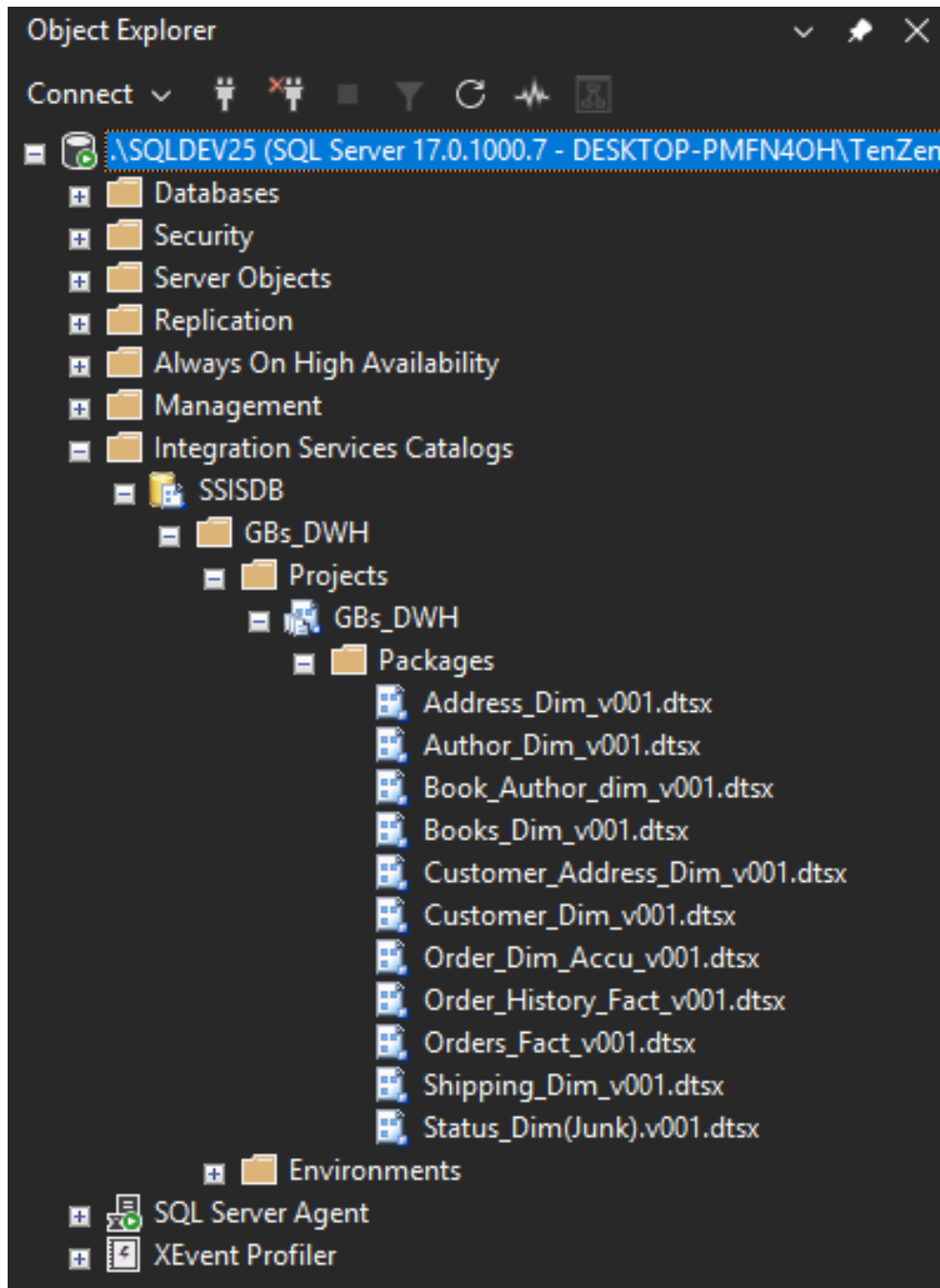
Order Fact Integration:



Order Fact History Integration:



Building the solution into SSMS:



The steps of the scheduling:

Job Properties - integrate data into GBs_DWH

Select a page: General, **Steps**, Schedules, Alerts, Notifications, Targets

Script Help

Job step list:

Step	Name	Type	On Success	On Failure
1	book_dim	SQL Server...	Go to the n...	Quit the job ...
2	author_dim	SQL Server...	Go to the n...	Quit the job ...
3	book_author_dim	SQL Server...	Go to the n...	Quit the job ...
4	shipping_dim	SQL Server...	Go to the n...	Quit the job ...
5	customer_dim	SQL Server...	Go to the n...	Quit the job ...
6	address_dim	SQL Server...	Go to the n...	Quit the job ...
7	customer_address_dim	SQL Server...	Go to the n...	Quit the job ...
8	orders_fact	SQL Server...	Go to the n...	Quit the job ...
9	order_dim	SQL Server...	Go to the n...	Quit the job ...
10	status_dim	SQL Server...	Go to the n...	Quit the job ...
11	order_history_fact	SQL Server...	Go to the n...	Quit the job ...
12	handle a problem in history fact	Transact-S...	Quit the job ...	Quit the job ...

Connection:

Server: DESKTOP-PMFN40H\SQLDEV2

Connection: DESKTOP-PMFN40H\TenZen9

[View connection properties](#)

Progress: Ready

Move step: Start step: 1:book_dim

New... Insert... Edit Delete

OK Cancel

Job Schedule Properties - GBs_DWH

Name: GBs_DWH Jobs in Schedule

Schedule type: Recuring ☒ Enabled

One-time occurrence

Date: 22/04/2026 Time: 5:45:10 PM

Frequency

Occurs: Daily

Recurs every: 1 day(s)

Daily frequency

☒ Occurs once at: 10:05:00 PM

☐ Occurs every: 1 hour(s) Starting at: 10:05:00 PM Ending at: 11:59:59 PM

Duration

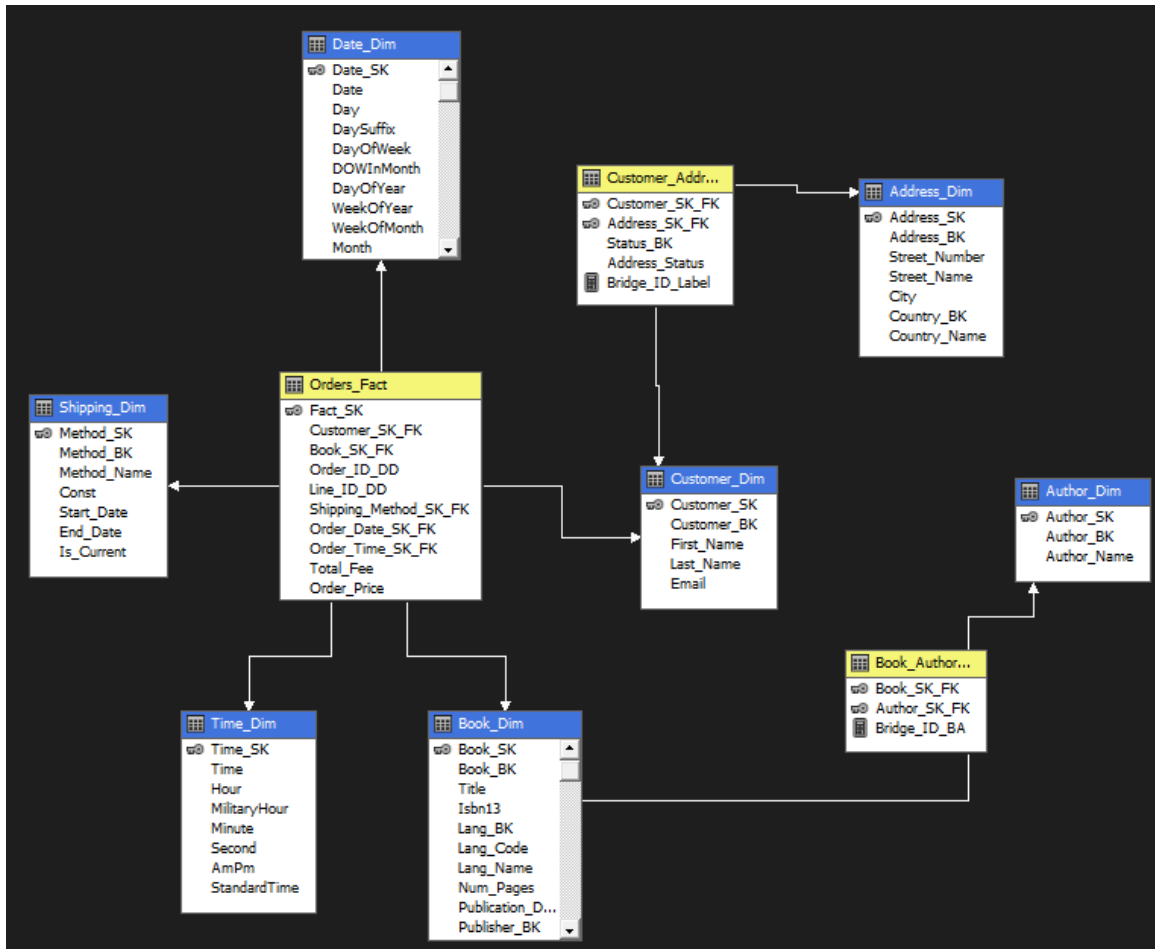
Start date: 18/04/2026 ☐ End date: 22/04/2026 ☒ No end date:

Summary

Description: Occurs every day at 10:05:00 PM. Schedule will be used starting on 22/04/2026.

OK Cancel Help

Building the cube with SSAS:



Creating the needed analysis by the business with Power BI:

